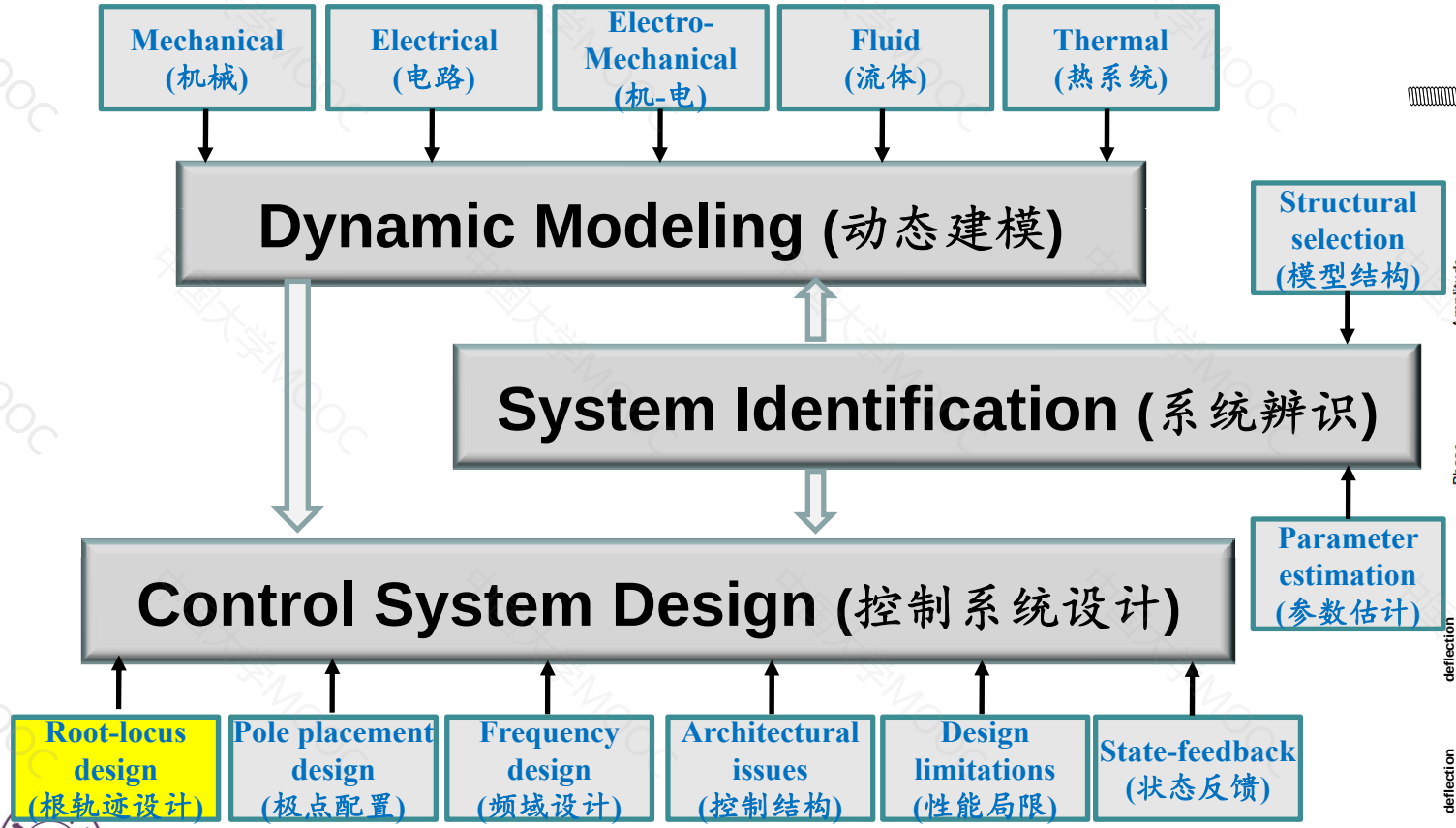
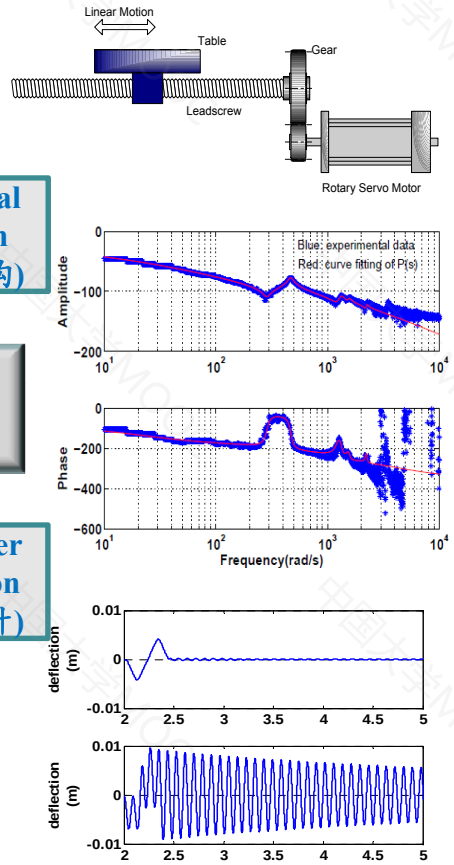
Controller Design via Root Locus – Principle

(基于根轨迹的控制器设计– 原理)

Main Contents of Course (课程主要内容)

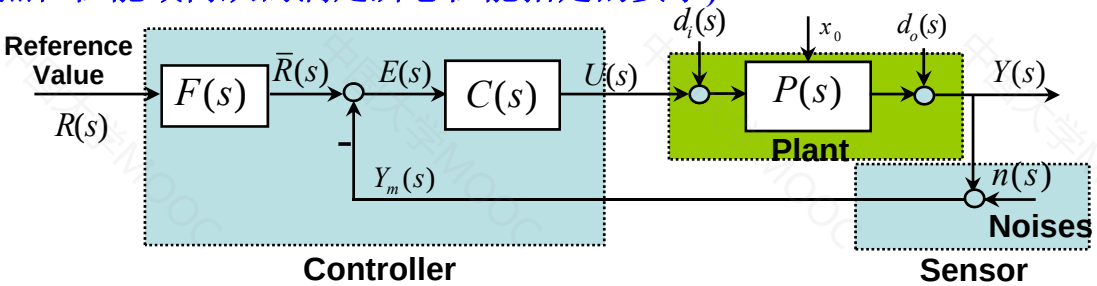




Principle of Controller Design via Root Locus

- **Underlying Design Principles** (设计原理)

  Choosing a feedback controller TF such that all the dominant CL poles are in the performance region to meet the transient performance specifications (选择一个反馈控制器传递函数使主导闭环极点在性能域内从而满足瞬态性能指定的要求):



$$P(s)=\frac{N_p(s)}{D_p(s)}=\frac{b_{pm_p}(s-z_{p1})\cdots(s-z_{pm_p})}{(s-p_{p1})\cdots(s-p_{pn_p})}$$

$$C(s)=K_c\frac{N_c(s)}{D_c(s)}=K_c\frac{(s-z_{c1})\cdots(s-z_{cm_c})}{(s-p_{c1})\cdots(s-p_{cn_c})}$$

Closed-Loop Poles (闭环极点):

$$\underbrace{D_p(s)D_c(s)}_{D(s)}+\underbrace{K_c}_{K}\underbrace{N_p(s)N_c(s)}_{N(s)}=0 \quad \text{or} \quad 1+K\frac{N(s)}{D(s)}=0 \Leftrightarrow 1+L(s)=0$$

Open-loop poles (开环极点): $D(s)=0 \quad \Leftrightarrow \quad \overbrace{D_p(s)}^{plant\ TF\ poles}=0 \quad or \quad \overbrace{D_c(s)}^{controller\ TF\ poles}=0$

Open-loop zeros (开环零点): $N(s)=0 \quad \Leftrightarrow \quad \overbrace{N_p(s)}^{plant\ zeros} \quad or \quad \overbrace{N_c(s)}^{controller\ TF\ zeros}=0$

Principle of Controller Design via Root Locus

⇒ **Role of Zeros of the Controller TF $C(s)$ (i.e., Roots of $N_c(s)$)** (控制器零点的作用):

Roots of $N_c(s)$ are also roots of $N(s)$ ($N_c(s)$的根也是$N(s)$的根) ⇒ **add open-loop zeros** (增加开环零点)

Role of Poles of the Controller TF $C(s)$ (i.e., Roots of $D_c(s)$) (控制器极点的作用):

Roots of $D_c(s)$ are also roots of $D(s)$ ($D_c(s)$的根也是$D(s)$的根) ⇒ **add open-loop poles**(增加开环极点)

- **Design Procedure**

Step1: Pick up $N_c(s)$ and $D_c(s)$ (i.e., choosing the additional open-loop zeros and poles that are needed) such that the resulting Root Locus intersects performance region or pass through the desired CL pole locations; Angle Criterion may be used in determining the locations of the needed additional open-loop poles and zeros (选取$N_c(s)$和$D_c(s)$(也就是选择需要增加的开环零点和极点)，以便使最终的根轨迹与性能域相交或经过期望的闭环极点位置。可用相角准则来确定所需增加的开环极点和零点的位置)

Step2: Pick up a gain K_c such that all CL poles are in the performance region or at the desired locations; Magnitude Criterion should be used to find the gain K_c corresponding to the desired CL poles (选取增益K_c，以便使所有的闭环极点在性能域内或在期望的位置。幅值准则应当被用来找到相应期望闭环极点的增益K_c)

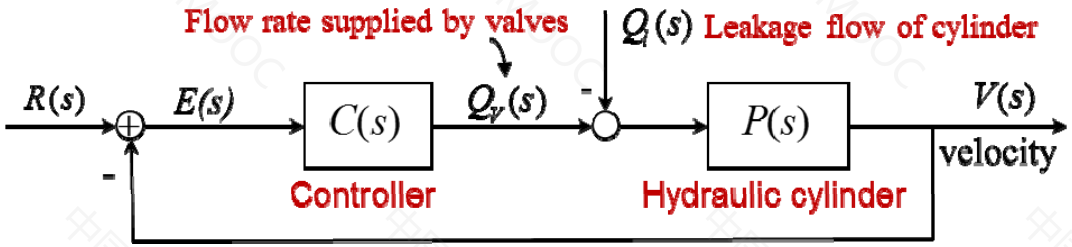
Ex.RL1 Motion Control of Hydraulic Cylinders

Hydraulic excavators exhibit severe chattering problem since the hydraulic cylinder dynamics is described by a lightly damped second-order system. You are required to design a feedback controller to solve this problem by satisfying the following transient performance requirements (因液压缸动态由欠阻尼二阶系统描述，液压挖掘机呈现严重振荡问题。请设计反馈控制器来解决此问题并满足下列瞬态性能要求):

(P1). The resulting CL system should have a OS% less than 4.3% (最终闭环系统的超调小于4.3%).

(P2). 2% settling time t_s should be smaller than 1 second (2%调整时间应当小于1s).

Block Diagram of Feedback Control System (反馈控制系统的方框图):



$$P(s)=\frac{K_p\omega_p^2}{s^2+2\zeta_p\omega_p s+\omega_p^2}=K_p\omega_p^2\frac{N_p(s)}{D_p(s)}$$

ω_p=3Hz=6π rad/s,

ζ_p=0.1, K_p=0.2

$$\Rightarrow D_p(s)=s^2+2\zeta_p\omega_p s+\omega_p^2=(s-p_1)(s-p_2),\quad p_{1,2}=-1.89\pm 18.79j$$

Performance Regions (性能域):

(P1) OS% ≤ 4.3%:

$$\Rightarrow \quad \varphi=-\tan^{-1}\left(\frac{\pi}{\log(OS\%100)}\right) \text{ or } \zeta\ge\sqrt{\frac{\left[\ln\left(\frac{X}{100}\right)\right]^2}{\pi^2+\left[\ln\left(\frac{X}{100}\right)\right]^2}}$$

$\Rightarrow \varphi = 45°$

or $\zeta = 0.707$

(P2) $t_s \le 1$ sec

$\Rightarrow$ τ ≤ ¼ sec $\Rightarrow$ σ ≥ 1/τ = 4

Ex.RL1 Motion Control of Hydraulic Cylinders

Proportional (P) Feedback Controller (比例反馈控制器):

In s-domain:

$$C(s)=K_c$$

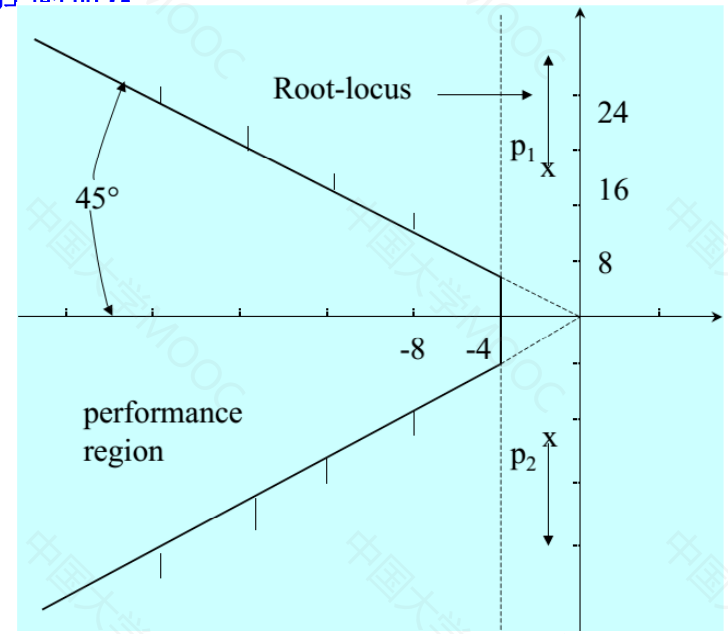
In time-domain:

$$q_V(t) = K_c\, e(t)$$

Root-Locus for P-Control:

$\Rightarrow$

$$\begin{cases} D(s)=D_p(s)D_c(s)=D_p(s)=(s-p_1)(s-p_2) \\ N(s)=N_p(s)N_c(s)=N_p(s)=1 \\ K=K_cK_p=71K_c \end{cases}$$



Q: Can we use a proportional (P) feedback controller to achieve our objective? (我们能用一个比例反馈控制器来达到我们的目标吗)

No. Since the root-locus does not intersect the performance region. (不能，因为根轨迹与性能域不相交)

Ex.RL1 Motion Control of Hydraulic Cylinders

Proportional plus Derivative (PD) feedback controller (比例微分控制器):

Control Law in Time-Domain:

$$q_V(t) = \underbrace{K_{CP}e(t)}_{\substack{\text{proportional}\\ \text{feedback}}} + \underbrace{K_{CD}\dot{e}(t)}_{\substack{\text{derivative}\\ \text{feedback}}} \quad \text{where} \quad \underbrace{e(t) = r(t) - y(t)}_{\text{tracking error}}$$

Control Law in s-Domain:

$$Q_V(s) = K_{CP}E(s) + K_{CD}sE(s) = (K_{CP} + K_{CD}s)\,E(s) = \underbrace{K_{CD}\left(s + \frac{K_{CP}}{K_{CD}}\right)}_{C(s)} E(s)$$

$$C(s) = K_{CD}\left(s + \frac{K_{CP}}{K_{CD}}\right) = K_C(s+\alpha)$$

where $K_C = K_{CD}$, $N_c(s) = s + \alpha$, $D_c(s) = 1$, $\alpha = \dfrac{K_{CP}}{K_{CD}}$

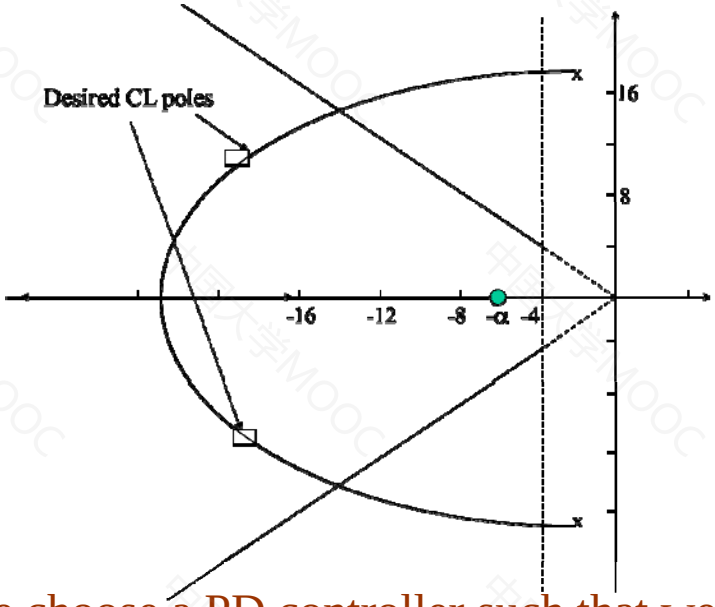
Closed-Loop Characteristic Equation (CLCE):

$D(s) = D_p(s)D_c(s) = D_p(s) = (s - p_1)(s - p_2)$ $\Rightarrow$ two open-loop poles at p_1 and p_2

$N(s) = N_p(s)N_c(s) = (s + \alpha)$ $\Rightarrow$ one open-loop zeros at $-\alpha$

$K = K_cK_p = 71K_c$ CLCE: $D(s) + KN(s) = 0$

Ex.RL1 Motion Control of Hydraulic Cylinders



Q: Can we choose a PD controller such that we can achieve our objective? (我们能选择一个PD控制器来实现我们的目标吗？)

Yes. As you can see from the sketch of root-locus, by placing the additional open-loop zero $z_1=-\alpha$ far away imaginary axis enough, the resulting root-locus will intersect the performance region (是的，从根轨迹可看出，通过将增加的开环零点放置在离虚轴足够远的位置，最终系统的根轨迹会和性能域相交).

Ex.RL1 Motion Control of Hydraulic Cylinders

Select Controller Gains **(选择控制器增益)**

For the above PD controller, choose controller gains so that the resulting CL poles are at -18 ± 12*j*, which are in the performance region **(采用上述PD控制器，选择控制器增益使闭环极点在性能域内的-18 ± 12*j*,**

Method 1: Direct Pole-Placement **(方法1：直接极点配置)**

Step 1: *Obtain the CLCE Expression* **(获得闭环特征方程表达式)**

$$D(s)+KN(s)=D_p(s)+K(s+\alpha)=s^2+3.77s+355+K(s+\alpha)$$
$$=s^2+(3.77+K)s+355+K\alpha$$

Step 2: *Obtain the Desired CLCE from Desired CL Pole Locations* **(由期望闭环极点位置获得期望的闭环特征方程)**

Want the desired CL poles at $p_{1,2d}$ = -18 ± 12*j*

$\Rightarrow$ Desired CLCE: $(s-p_{1d})(s-p_{2d})=(s+18-12j)(s+18+12j)=s^2+36s+468$

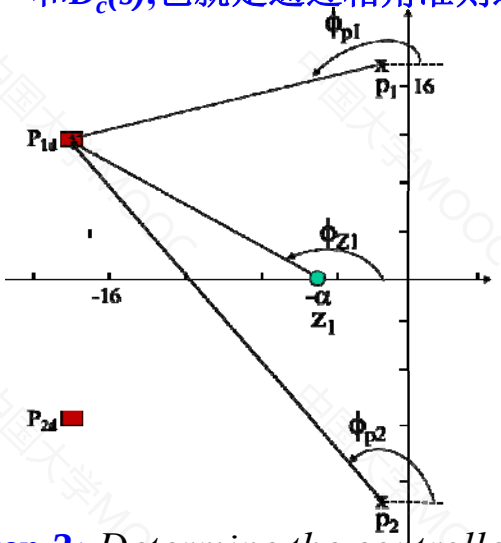
Step 3: *Match the CLCE expression with the desired CLCE to obtain a set of equations for solving unknown controller gains* **(使闭环特征方程表达式与期望特征方程相匹配，获得一系列方程求解得出未知控制器增益)**

$$\begin{cases}3.77+K=36\\355+K\alpha=468\end{cases}\Rightarrow\begin{cases}K=32.2\\\alpha=3.5\end{cases}\Rightarrow K_{CD}=K_c=\frac{K}{K_p}=\frac{32.2}{71}=0.45,K_{CP}=K_{CD}\alpha=1.59$$

Ex.RL1 Motion Control of Hydraulic Cylinders

Method 2: Pole-Placement via Root Locus **(方法2：通过根轨迹的极点配置)**

Step 1: *Pick up compensator $N_c(s)$ and $D_c(s)$, i.e., choose the additional open-loop poles/zeros so that the resulting root locus pass the desired CL pole locations via Angle Criterion* **(选择补偿器$N_c(s)$和$D_c(s)$,也就是通过相角准则选择增加的开环零/极点以便最终的根轨迹经过期望的闭环极点位置):**



$$\phi_{p_1} = \angle(p_{1d} - p_1) = -157^\circ \qquad \phi_{p_2} = \angle(p_{1d} - p_2) = 117.7^\circ$$

$$\phi_{z_1} = \angle(-18 + \alpha + 12j) = 180^\circ - \tan^{-1}\left(\frac{12}{18-\alpha}\right) \qquad \text{(Assume } \alpha < 18\text{)}$$

Angle Criterion**(相角准则):**

$$\angle(p_{1d} - z_1) - \angle(p_{1d} - p_1) - \angle(p_{1d} - p_2) = 180^\circ$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

$$\phi_{z1} \qquad \phi_{p1} \qquad \phi_{p2}$$

$$\Rightarrow \phi_{z1} = 180^\circ + \phi_{p1} + \phi_{p2} = 140.04^\circ$$

$$\frac{12}{-18+\alpha} = \tan(\phi_{z1} - 180^\circ) = -0.827$$

$$\Rightarrow \alpha = 3.5$$

Step 2: *Determine the controller gain K_c via Magnitude Condition* **(通过幅值准则确定控制器增益K_c)**

Magnitude Condition **(幅值准则):** $K = \dfrac{|p_{1d} - p_1| \cdot |p_{1d} - p_2|}{|p_{1d} - z_1|} = \dfrac{7.5 \times 34}{18.8} = 32.2$

Ex.RL2 Motion Control of Hydraulic Cylinders

Let us reconsider Ex.RL1. In addition to the transient performance requirements (P1) and (P2), the CL system is required to meet the following steady-state (SS) performance requirement **(考虑Ex.PL1,除了瞬态性能要求(P1)和(P2)之外，闭环系统要求满足下列稳态性能要求)**:

(P3). Zero steady-state tracking error for any constant reference velocity command **(对任意常值参考速度输入具有零稳态跟踪误差).**

Q: Can the previously designed PD controller meet the SS requirement? **(上述设计的PD控制器满足稳态性能要求吗？)**

$$T_r(s) = \frac{KN(s)}{D(s)+KN(s)} = \frac{32.2\times(s+3.5)}{s^2+36s+468}$$

since

$$\lim_{s\to 0}\frac{1-T_r(s)}{s^0} = 1-T_r(s) = 1-\frac{32.2\times 3.5}{468} = 0.76$$

The resulting CL system is of type 0 w.r.t. reference input and the steady-state tracking error for a unit step reference input is 0.76. Therefore, PD controller can NOT meet the SS requirement **(最终的闭环系统对参考输入是零型系统，对单位阶跃参考输入的稳态跟踪误差是0.76。因此，PD控制器不能满足稳态性能要求).**

Ex.RL2 Motion Control of Hydraulic Cylinders

Proportional plus Derivative plus Integral (PID) feedback controller:

Control Law in Time-Domain (时域控制律):

$$q_V(t) = \underbrace{K_{CP}e(t)}_{\substack{\text{proportional}\\ \text{feedback}}} + \underbrace{K_{CD}\dot{e}(t)}_{\substack{\text{derivative}\\ \text{feedback}}} + \underbrace{K_{CI}\int e(t)dt}_{\substack{\text{integral}\\ \text{feedback}}}$$

$$T_r(s) = \frac{KN(s)}{D(s)+KN(s)} = \frac{K_cK_pN_p(s)N_c(s)}{D_p(s)D_c(s)+K_pK_cN_p(s)N_c(s)}$$

$$= \frac{K(s-z_1)(s-z_2)}{s(s-p_1)(s-p_2)+K(s-z_1)(s-z_2)}$$

$$1-T_r(s) = \frac{s(s-p_1)(s-p_2)}{s(s-p_1)(s-p_2)+K(s-z_1)(s-z_2)}$$

Control Law in s-Domain (频域控制律):

$$Q_V(s) = K_{CP}\,E(s) + K_{CD}\,sE(s) + K_{CI}\cdot\frac{1}{s}E(s) = \underbrace{(K_{CP} + K_{CD}\,s + K_{CI}\cdot\frac{1}{s})}_{C(s)}E(s)$$

$$C(s) = K_{CP} + K_{CD}\,s + \frac{1}{s}K_{CI} = \frac{K_{CD}\,s^2 + K_{CP}\,s + K_{CI}}{s} = \frac{K_c(s-z_{c1})(s-z_{c2})}{s}$$

$$\begin{aligned} K_{CD} &= K_c \\ K_{CI} &= K_{CD}\,z_{c1}z_{c2} \\ K_{CP} &= -K_{CD}(z_{c1}+z_{c2}) \end{aligned} \Leftrightarrow \begin{cases} K_c = K_{CD} \\ N_c(s) = (s-z_{c1})(s-z_{c2}) \\ D_c(s) = s \end{cases}$$

PID controller adds one additional pole at $p_{c1} = 0$ and two additional open-loop zeros at z_{c1} and z_{c2} **(PID控制器增加了一个极点$p_{c1} = 0$和2个开环零点z_{c1} 和 z_{c2})**

***Q: Can PID controller meet the steady-state requirement P3?*(PID控制器能满足稳态性能要求(P3)吗?)**

Yes. Since $\lim_{s\to 0}\{1-T_r(s)\} = \lim_{s\to 0}\frac{sp_1p_2}{Kz_1z_2} = 0 \Longrightarrow$ The CL system is of type 1 w.r.t. reference input **(闭环系统对参考输入是类型1的)**

For any constant reference input, SS error is zero **(对任意常值参考输入，稳态误差是零)**

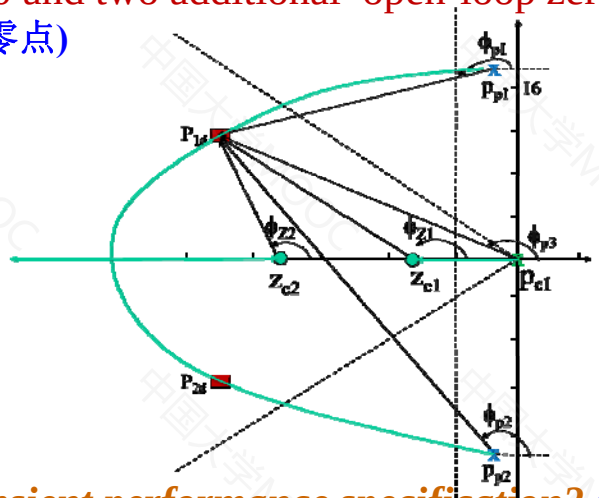
Ex.RL2 Motion Control of Hydraulic Cylinders

Closed-Loop Characteristic Equation(CLCE) **(闭环特征方程):**

$$D(s)+KN(s)=0 \text{ or } \underbrace{s(s-p_1)(s-p_2)}_{D(s)}+\underbrace{K(s-z_1)(s-z_2)}_{N(s)}=0$$

i.e., PID controller adds one additional pole at p_2= 0 and two additional open-loop zeros at z_1 and z_2 **(PID控制器增加了一个额外的极点和两个额外的开环零点)**

Root-Locus for PID-Control **(PID控制的根轨迹):**



Q: Can we choose a PID controller to meet the transient performance specification? **(我们能选择一个PID控制器来满足瞬态性能要求吗?)**

Yes. Since the root-locus will intersect performance region as shown in above **(是的，因为根轨迹与性能域相交)**

Ex.RL2 Motion Control of Hydraulic Cylinders

Select PID Gains: **(选择PID增益)**

For the above PID controller, choose controller gains so that the dominant poles of the resulting CL system are at -18 ± 12*j*, which are in the performance region **(对上述PID控制器，选择控制器增益以便闭环系统主导极点在性能域内的点-18 ± 12*j*)**

Method 1: Direct Pole-Placement **(直接极点配置)**

Step 1: *Obtain the CLCE Expression***(获得闭环特征方程表达式)**

$$D(s)+KN(s)=s(s-p_1)(s-p_2)+K(s-z_1)(s-z_2)=s^3+3.77s^2+355s+K[s^2-(z_1+z_2)s+z_1z_2]$$

$$=s^3+(3.77+K)s^2+[355-K(z_1+z_2)]s+Kz_1z_2$$

Step 2: *Obtain the Desired CLCE from Desired CL Pole Locations* **(由期望闭环极点获得期望特征方程)**

$p_{1,2d}=-18\pm 12j$, Choose K, z_1, z_2 to place the three CL poles at any position as we want (p_{3d}= -20) $\Rightarrow$ Desired CLCE: $(s-p_{1d})(s-p_{2d})(s-p_{3d})=(s+18-12j)(s+18+12j)(s+20)$

$$=s^3+56s^2+1188s+9360$$

Step 3: *Match the CLCE expression with the desired CLCE to obtain a set of equations for solving unknown controller gains* **(匹配特征方程式和期望特征方程获得方程求解得到未知控制器增益)**

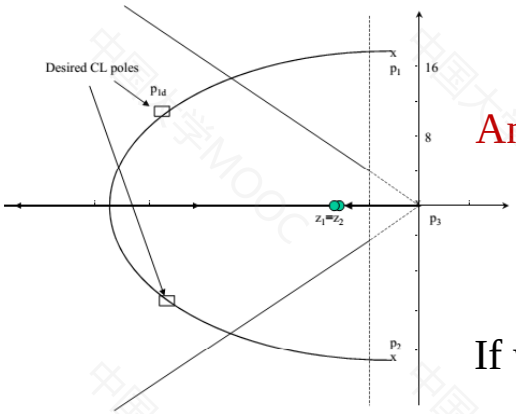
$$\begin{cases}3.77+K=56\\355-K(z_1+z_2)=1188\\Kz_1z_2=9360\end{cases}\Rightarrow\begin{cases}K=52.23\\z_1=-8+10.8j\\z_2=-8-10.8j\end{cases}$$

$$\Rightarrow K_{CD}=K_c=\frac{K}{K_p}=\frac{52.23}{71}=0.736,K_{CP}=-K_{CD}(z_1+z_2)=11.73,K_{CI}=K_{CD}z_1z_2=131.8$$

Ex.RL2 Motion Control of Hydraulic Cylinders

Method 2: Pole-Placement via Root Locus **(通过根轨迹的极点配置)**

Step 1: *Pick up compensator $N_c(s)$ and $D_c(s)$, i.e., choose the additional open-loop poles/zeros so that the resulting root locus pass the desired CL pole locations via Angle Criterion:* **(通过相角准则选择补偿器,也就是选择增加的开环极点/零点以便最终根轨迹经过期望的闭环极点)**



$$\phi_{p_1} = \angle(p_{1d} - p_1) = -157^\circ \qquad \phi_{p_2} = \angle(p_{1d} - p_2) = 117.7^\circ \qquad \phi_{p_3} = \angle(p_{1d} - p_3) = 146.3^\circ$$

$$\phi_{z_1} = \angle(-18 + z_1 + 12j) = 180^\circ - \tan^{-1}\left(\frac{12}{18-\alpha}\right)$$

Angle Criterion **(相角准则)**:

$$\underbrace{\angle(p_{1d} - z_1)}_{\phi_{z1}} + \underbrace{\angle(p_{1d} - z_2)}_{\phi_{z2}} - \underbrace{\angle(p_{1d} - p_1)}_{\phi_{p1}} - \underbrace{\angle(p_{1d} - p_2)}_{\phi_{p2}} - \underbrace{\angle(p_{1d} - p_3)}_{\phi_{p3}} = 180^\circ$$

$$\Rightarrow \phi_{z1} + \phi_{z2} = 180^\circ + \phi_{p1} + \phi_{p2} + \phi_{p3} = 286.7^\circ$$

Case A: Adding a pair of real zeros **(增加一对实零点)**

If we choose **(如果选择)**

$$z_1 = z_2, \text{ then } \phi_{z1} = \phi_{z2} = 143.4^\circ \quad \Rightarrow z_1 = z_2 = -18 - \frac{12}{\tan(\phi_{z1} - 180^\circ)} = -1.86$$

The resulting CL system **(最终闭环环系统)** $p_{3d} = -0.24,\ p_{1d} = -18 + 12j,\ p_{2d} = -18 - 12j$

may not be able to satisfy setting time requirements **(可能不能满足调整时间要求)**

→ not a good design! **(不是好的设计)**

Ex.RL2 Motion Control of Hydraulic Cylinders

Method 2: Pole-Placement via Root Locus **(通过根轨迹的极点配置)**

Step 2: *Determine the controller gain K_c via Magnitude Condition* **(通过幅值条件确定控制器增益)**

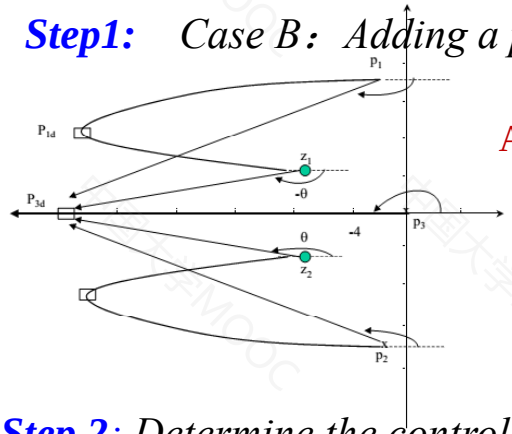
Magnitude Condition **(幅值准则)** for $\boldsymbol{p_{1d}}$:

$$K = \frac{|p_{1d} - p_1| \cdot |p_{1d} - p_2||p_{1d} - 0|}{|p_{1d} - z_1||p_{1d} - z_2|} = \frac{17.47 \times 34.7 \times 21.6}{20.1^2} = 32.4$$

Ex.RL2 Motion Control of Hydraulic Cylinders

Method 2: Pole-Placement via Root Locus **(通过根轨迹的极点配置)**

Step1: *Case B：Adding a pair of complex zeros* **(增加一对共轭复数零点)**



$$z_1 = -a + bj \text{ and } z_2 = -a - bj$$

Angle Criterion **(相角准则):**

$$\angle(p_{3d} - z_1) + \angle(p_{3d} - z_2) - \angle(p_{3d} - p_1) - \angle(p_{3d} - p_2) - \angle(p_{3d} - p_3) = 180^\circ$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$-\theta \quad \theta \quad -134^\circ \quad 134^\circ \quad 180^\circ$$

Always satisfy Angle Condition!**(始终满足相角条件)**

Step 2*: Determine the controller gain K_c via Magnitude Condition* **(通过幅值条件确定控制器增益)**

Magnitude Condition **(幅值条件)** for $\boldsymbol{p_{1d}}$:

$$K = \frac{|p_{1d} - p_1| \cdot |p_{1d} - p_2| |p_{1d} - p_3|}{|p_{1d} - z_1| |p_{1d} - z_2|} = \frac{17.47 \times 34.7 \times 21.6}{\sqrt{(-18+a)^2 + (12-b)^2}\sqrt{(-18+a)^2 + (12+b)^2}}$$

Magnitude Condition **(幅值条件)** for $\boldsymbol{p_{3d}}$ **=-20**:

$$K = \frac{|p_{3d} - p_1| \cdot |p_{3d} - p_2| |p_{3d} - p_3|}{|p_{3d} - z_1| |p_{3d} - z_2|} = \frac{26 \times 26 \times 20}{\left(\sqrt{(-20+a)^2 + b^2}\right)^2}$$

$$\Rightarrow \begin{cases} a = 8 \\ b = 10.8 \\ K = 52.2 \end{cases} \text{ or } \begin{cases} z_{1,2} = -8 \pm 10.8j \\ K = 52.2 \end{cases}$$

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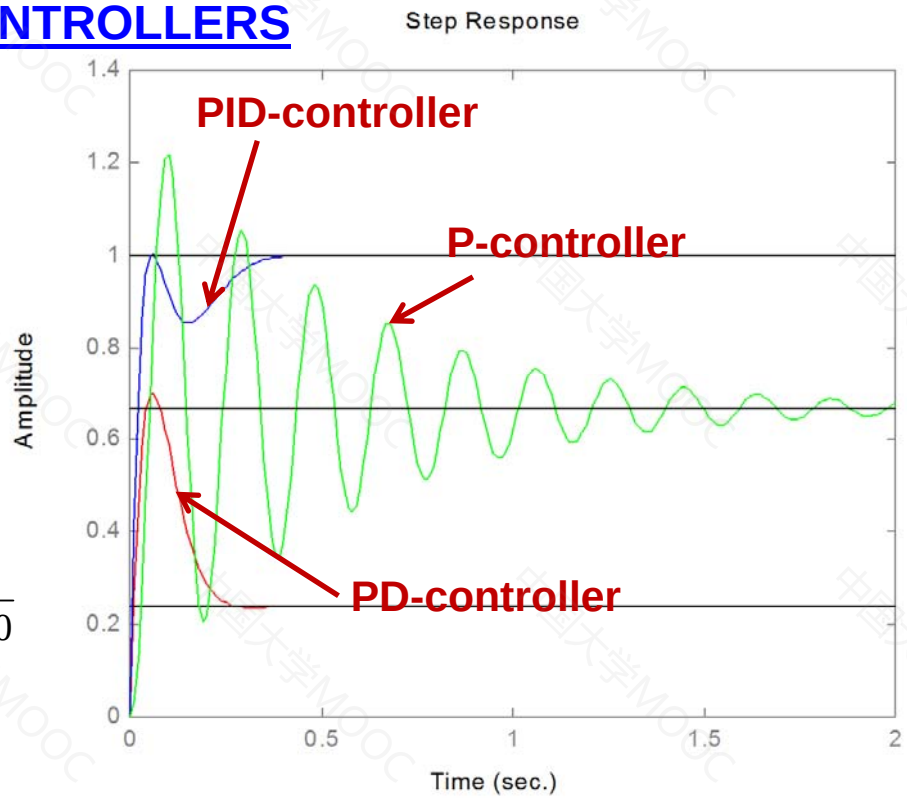
COMPARISON OF P, PD, and PID CONTROLLERS

The closed-loop transfer functions for different controllers are (不同控制器的闭环传递函数):

P-controller with K_c =10: $$T(s)=\frac{710}{s^2+3.77s+1065}$$

PD-controller: $$T(s)=\frac{32.2(s+3.5)}{s^2+36s+468}$$

PID-controller: $$T(s)=\frac{52.23s^2+833s+9360}{s^3+56s^2+1188s+9360}$$



Unit step responses of the P-controller, PD controller, and PID controller

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SUMMARY OF PID CONTROL

(a) The integral control is to reduce or eliminate constant steady-state errors, but this benefit typically comes at the cost of a degraded transient response if the controller is not well designed **(积分控制用来降低或消除常值稳态误差，但如果控制器没有设计好，瞬态响应会变差).**

(b) The derivative feedback is used in conjunction with proportional and/or integral feedback to increase damping and generally improve the stability and the transient performance of a system **(微分反馈用来连接比例和/或积分反馈以增加阻尼，从而提高系统瞬态性能的稳定性).**

Summary

- **Principle of Controller Design via Root Locus** (采用根轨迹的控制器设计原理)
  - ✓ **Adding zeros and poles of design controller to make the root-locus intersect performance region** (增加控制器零点和极点使根轨迹与期望性能域相交)
  - ✓ **P Controller**(比例控制器)
  - ✓ **PD Controller**(比例微分控制器)
  - ✓ **PID Controller**(比例积分微分控制器)